UNIT 4 · 9 hours

DC Load Line Analysis and Q-Point



Transistors

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Derive the DC load line for a BJT and a MOSFET stage
- Locate the Q-point
- Check the region of operation
- Choose the Q-point for amplifier or switch operation

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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DC load line equations

BJT: $V_{CE} = V_{CC} - I_C R_C$

output loop KVL

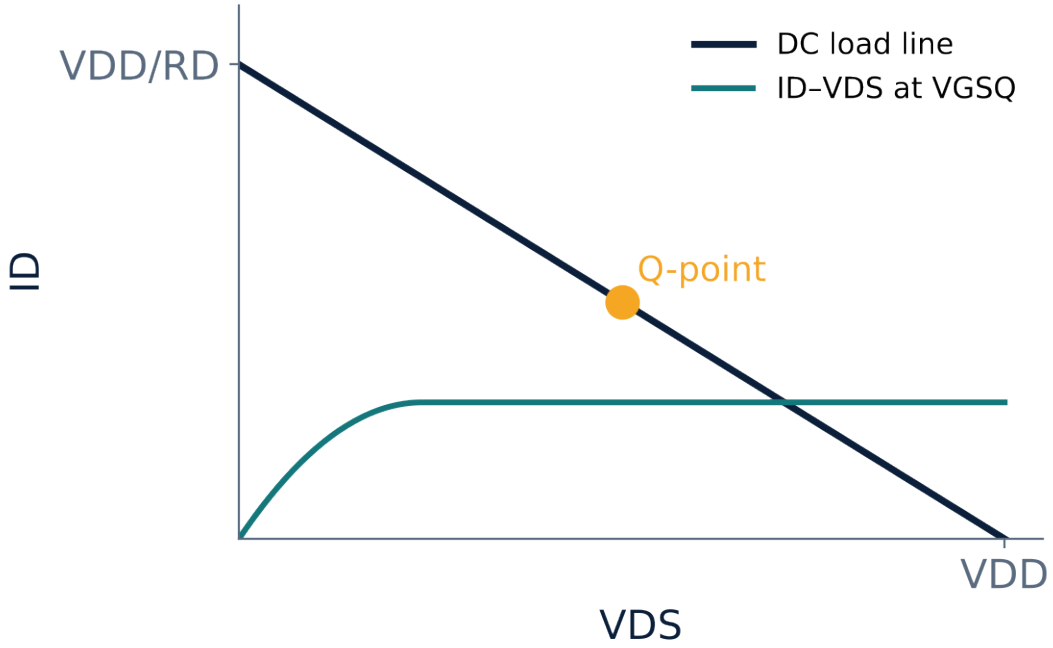
MOSFET: $V_{DS} = V_{DD} - I_D R_D$

output loop KVL

- Endpoints: $I_{C(max)} = V_{CC}/R_C$ at $V_{CE} = 0$, and $V_{CE} = V_{CC}$ at $I_C = 0$.
- Line of all possible (V_{CE}, I_C) pairs allowed by the external circuit.
- The device characteristic fixes the actual point on the line: the Q-point.

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Q-point on the load line (MOSFET)



- Intersection of the DC load line with the device curve for VGSQ.
- Q-point = (VDSQ, IDQ).
- For an amplifier: put Q near the middle of the line to allow equal swing.
- Too high: clipping in cut-off; too low: clipping in saturation.

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Where to place the Q-point

Amplifier

Middle of the active/saturation region: maximum undistorted swing.

Switch OFF

At the cut-off end: IC ≈ 0, VCE ≈ VCC.

Switch ON

At the saturation end: IC ≈ VCC/RC, VCE ≈ 0.

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Numerical: MOSFET load line

Problem VDD = 12 V, RD = 2 kΩ, n-MOSFET (k = 0.5 mA/V², VT = 2 V) with VGS = 4 V. Find the load line endpoints and the Q-point.

- Endpoints: VDS = 12 V (ID = 0) and ID = 12/2k = 6 mA (VDS = 0)
- Assume saturation: ID = k(VGS − VT)² = 0.5 × 4 = 2 mA
- VDS = 12 − 2 mA × 2 kΩ = 8 V
- Check: VDS = 8 V ≥ VGS − VT = 2 V → saturation ✓

Answer: Q-point: ID = 2 mA, VDS = 8 V

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Numerical: BJT load line

Problem VCC = 12 V, RC = 3 kΩ, β = 100, IB = 20 µA. Find Q-point and check region.

- IC = β IB = 100 × 20 µA = 2 mA
- VCE = 12 − 2 mA × 3 kΩ = 6 V
- Endpoints: IC(max) = 4 mA, VCE(max) = 12 V → Q is at the middle

Answer: Q: IC = 2 mA, VCE = 6 V (active region)

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Key takeaways

- Load line from KVL: VCE = VCC − IC RC (or VDS = VDD − ID RD)
- Q-point = intersection of load line and device characteristic
- Q mid-line for amplifiers; ends of the line for switches
- Verify the region of operation

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Practice questions

Q1. Draw the DC load line for VCC = 10 V, RC = 2 kΩ.

Q2. Find the Q-point for VDD = 15 V, RD = 3 kΩ, ID = 2 mA.

Q3. What happens if the Q-point is too close to cut-off?

Q4. How does the load line differ for switch and amplifier use?